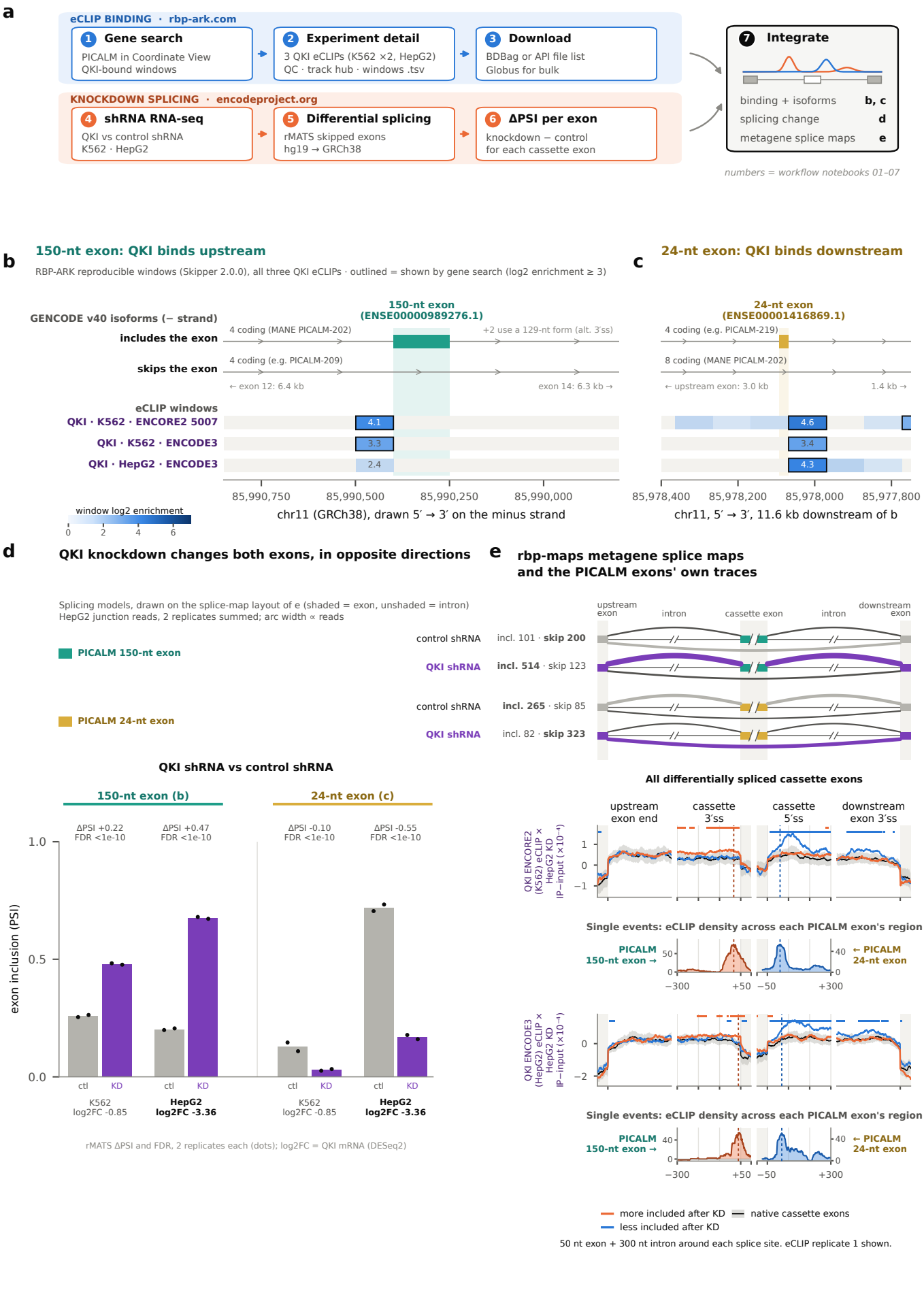

a
eCLIP BINDING · rbp-ark.com
1 Gene search
PICALM in Coordinate View
QKI-bound windows
2 Experiment detail
3 QKI eCLIPs (K562 ×2, HepG2)
QC · track hub · windows .tsv
3 Download
BDBag or API file list
Globus for bulk
KNOCKDOWN SPLICING · encodeproject.org
4 shRNA RNA-seq
QKI vs control shRNA
K562 · HepG2
5 Differential splicing
rMATS skipped exons
hg19 → GRCh38
6 ΔPSI per exon
knockdown − control
for each cassette exon
7 Integrate
binding + isoforms b, c
splicing change d
metagene splice maps e
numbers = workflow notebooks 01–07
b
150-nt exon: QKI binds upstream
RBP-ARK reproducible windows (Skipper 2.0.0), all three QKI eCLIPs · outlined = shown by gene search (log2 enrichment ≥ 3)
150-nt exon (ENSE00000989276.1)
GENCODE v40 isoforms (− strand)
includes the exon
4 coding (MANE PICALM-202)
+2 use a 129-nt form (alt. 3′ss)
skips the exon
4 coding (e.g. PICALM-209)
← exon 12: 6.4 kb
exon 14: 6.3 kb →
eCLIP windows
QKI · K562 · ENCORE2 5007
QKI · K562 · ENCODE3
QKI · HepG2 · ENCODE3
4.1
3.3
2.4
85,990,750
85,990,500
85,990,250
85,990,000
window log2 enrichment
0
2
4
6
chr11 (GRCh38), drawn 5′ → 3′ on the minus strand
c
24-nt exon: QKI binds downstream
24-nt exon (ENSE00001416869.1)
4 coding (e.g. PICALM-219)
8 coding (MANE PICALM-202)
← upstream exon: 3.0 kb
1.4 kb →
4.6
3.4
4.3
85,978,400
85,978,200
85,978,000
85,977,800
chr11, 5′ → 3′, 11.6 kb downstream of b
d
QKI knockdown changes both exons, in opposite directions
Splicing models, drawn on the splice-map layout of e (shaded = exon, unshaded = intron)
HepG2 junction reads, 2 replicates summed; arc width ∝ reads
PICALM 150-nt exon
PICALM 24-nt exon
control shRNA
incl. 101 · skip 200
QKI shRNA
incl. 514 · skip 123
control shRNA
incl. 265 · skip 85
QKI shRNA
incl. 82 · skip 323
QKI shRNA vs control shRNA
150-nt exon (b)
24-nt exon (c)
ΔPSI +0.22
FDR <1e-10
ΔPSI +0.47
FDR <1e-10
ΔPSI -0.10
FDR <1e-10
ΔPSI -0.55
FDR <1e-10
exon inclusion (PSI)
1.0
0.5
0.0
ctl
KD
K562
log2FC -0.85
HepG2
log2FC -3.36
rMATS ΔPSI and FDR, 2 replicates each (dots); log2FC = QKI mRNA (DESeq2)
e
rbp-maps metagene splice maps
and the PICALM exons' own traces
upstream exon
intron
cassette exon
intron
downstream exon
All differentially spliced cassette exons
upstream exon end
cassette 3′ss
cassette 5′ss
downstream exon 3′ss
QKI ENCORE2 (K562) eCLIP × HepG2 KD IP−input (×10⁻⁴)
Single events: eCLIP density across each PICALM exon's region
PICALM 150-nt exon →
← PICALM 24-nt exon
−300
+50
−50
+300
QKI ENCODE3 (HepG2) eCLIP × HepG2 KD IP−input (×10⁻⁴)
more included after KD
less included after KD
native cassette exons
50 nt exon + 300 nt intron around each splice site. eCLIP replicate 1 shown.